Hierarchical OSPF

- *two-level hierarchy:* local area, backbone.
  - link-state advertisements only in area
  - each nodes has detailed area topology; only know direction (shortest path) to nets in other areas.
- *area border routers:* "summarize" distances to nets in own area, advertise to other Area Border routers.
- *backbone routers:* run OSPF routing limited to backbone.
- *boundary routers:* connect to other AS' es.

Chapter 5: outline

5.1 introduction

5.2 routing protocols

- link state
- distance vector

5.3 intra-AS routing in the Internet:

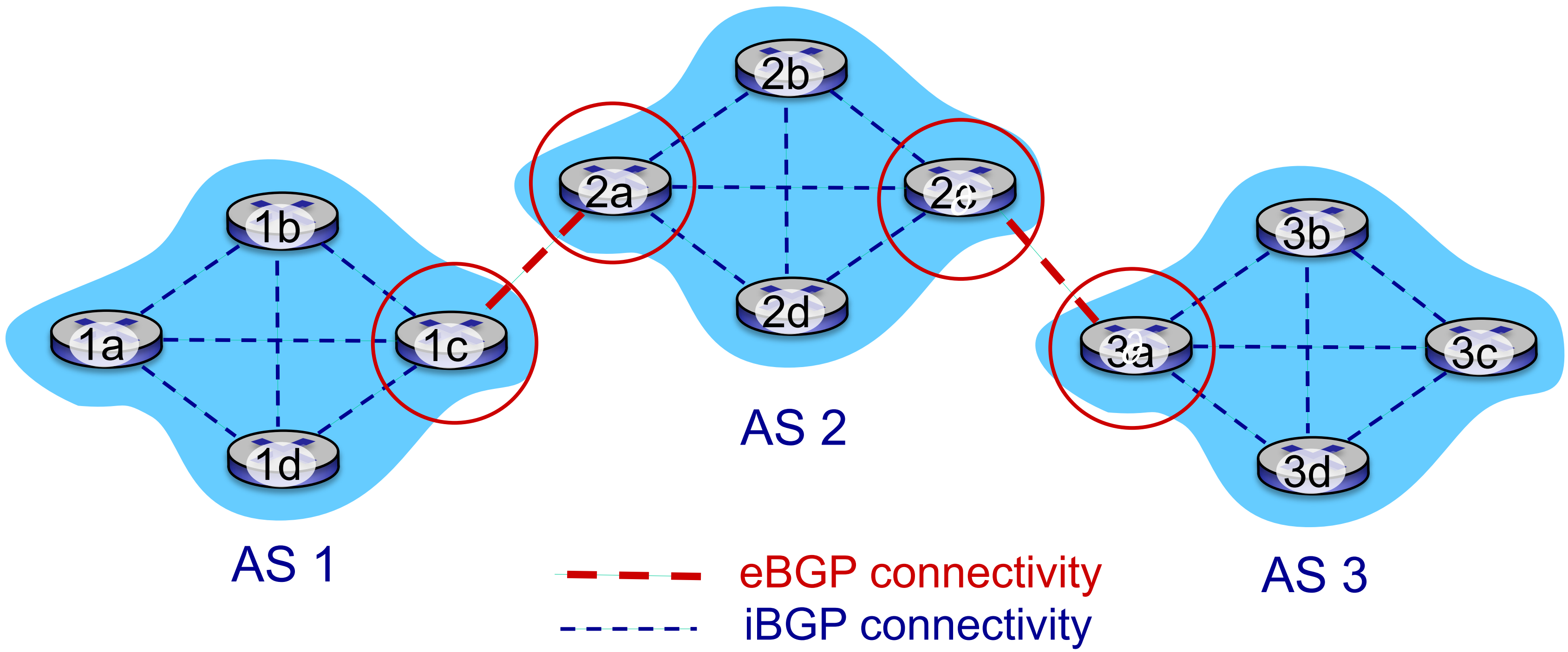
- RIP
- OSPF

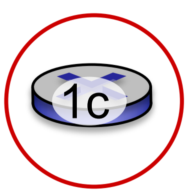
5.4 inter-AS routing in the Internet BGP

Internet inter-AS routing: BGP

- BGP (Border Gateway Protocol): *the* de facto inter-domain routing protocol (v 4)
  - "glue that holds the Internet together"
- BGP provides each AS a means to:
  - eBGP: obtain subnet reachability information from neighboring ASes
  - iBGP: propagate reachability information to all AS-internal routers.
  - determine "good" routes to other networks based on reachability information and *policy*
- allows subnet to advertise its existence to rest of Internet: *"I am here"*
- uses TCP for reliable communications to transmit routing messages

eBGP, iBGP connections

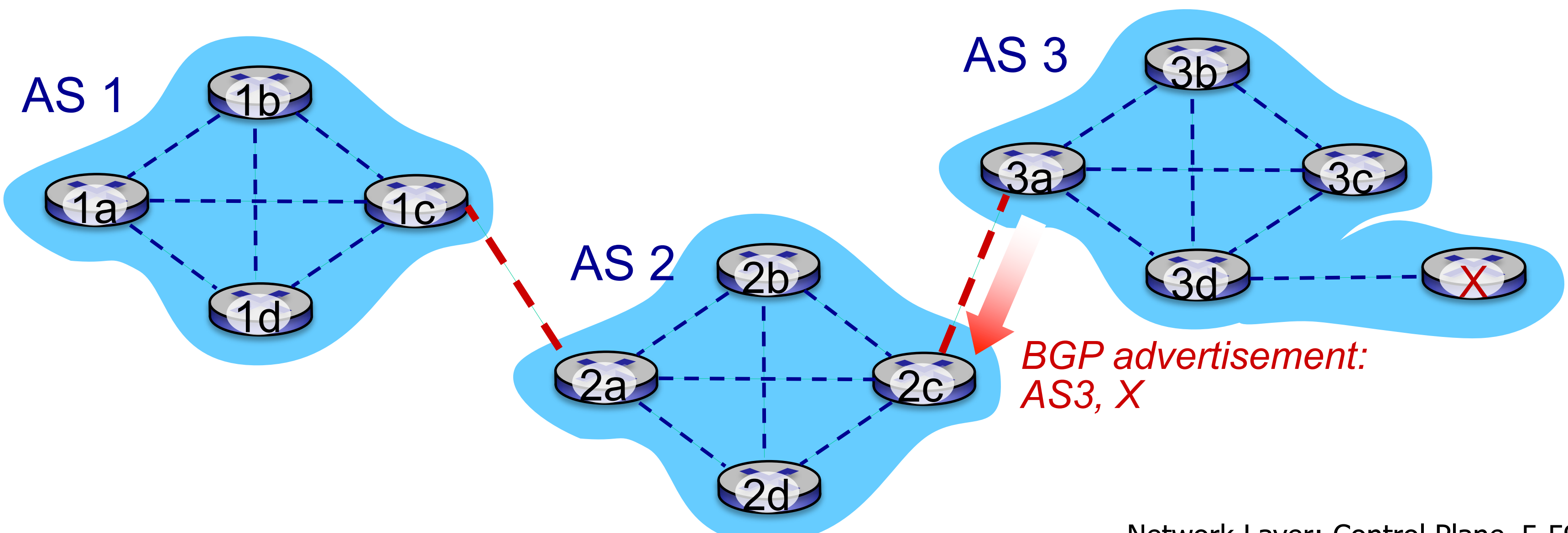




gateway routers run both eBGP and iBGP protocols

BGP basics

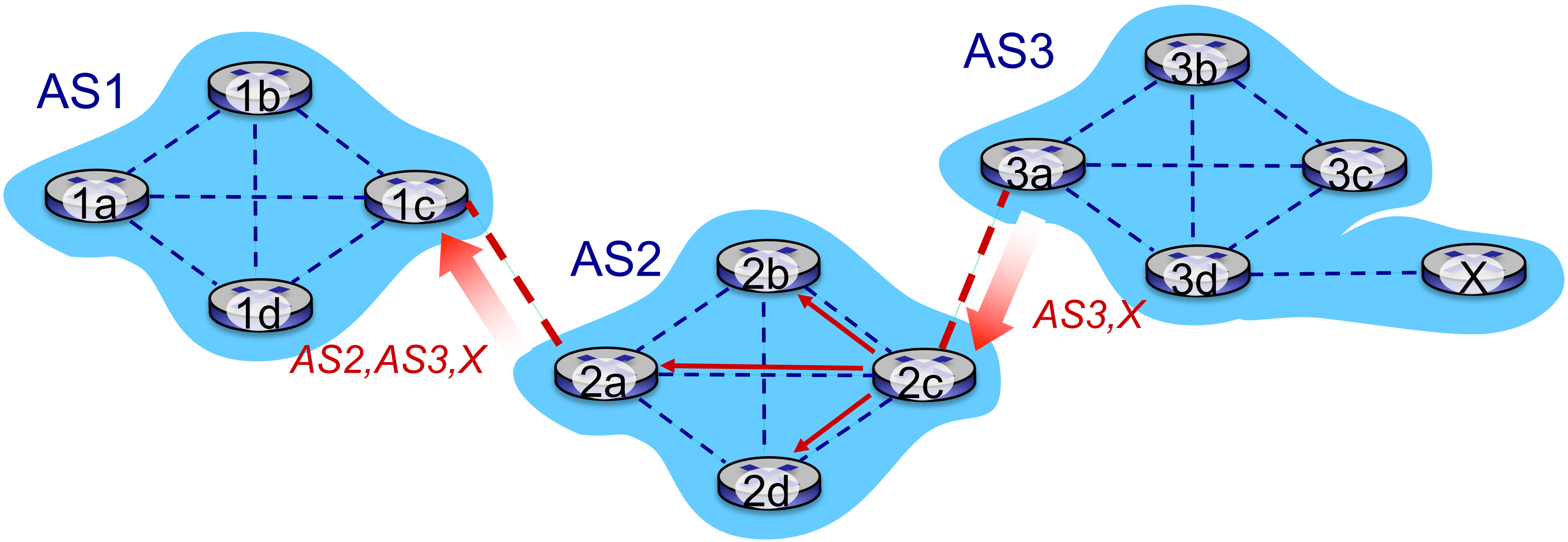
- BGP session: two BGP routers ("peers") exchange BGP messages over semi-permanent TCP connection:
  - advertising *paths* to different destination network prefixes (BGP is a "path vector" protocol)
- when AS3 gateway router 3a advertises path AS3,X to AS2 gateway router 2c:
  - AS3 *promises* to AS2 it will forward datagrams towards X



Path attributes and BGP routes

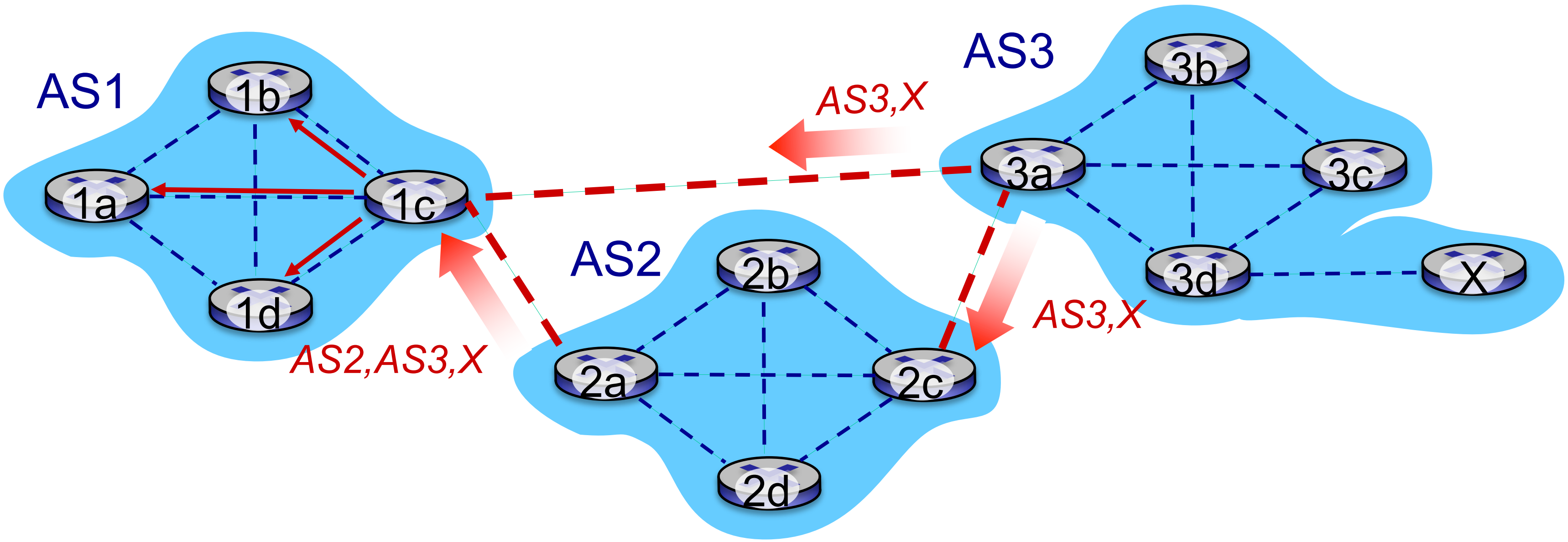
- advertised prefix includes BGP attributes
  - prefix + attributes = "route"
- two important attributes:
  - AS-PATH: list of ASes through which prefix advertisement has passed
  - NEXT-HOP: indicates specific internal-AS router to next-hop AS
- *Policy-based routing:*
  - gateway receiving route advertisement uses *import policy* to accept/decline path (e.g., never route through AS Y).
  - AS *export policy* also determines whether to *advertise* path to other neighboring ASes

BGP path advertisement



- AS2 router 2c receives path advertisement AS3,X (via eBGP) from AS3 router 3a
- Based on AS2 *import policy*, AS2 router 2c accepts path AS3,X, and propagates (via iBGP) to all AS2 routers
- Based on AS2 *export policy*, AS2 router 2a advertises (via eBGP) path AS2, AS3, X to AS1 router 1c

BGP path advertisement

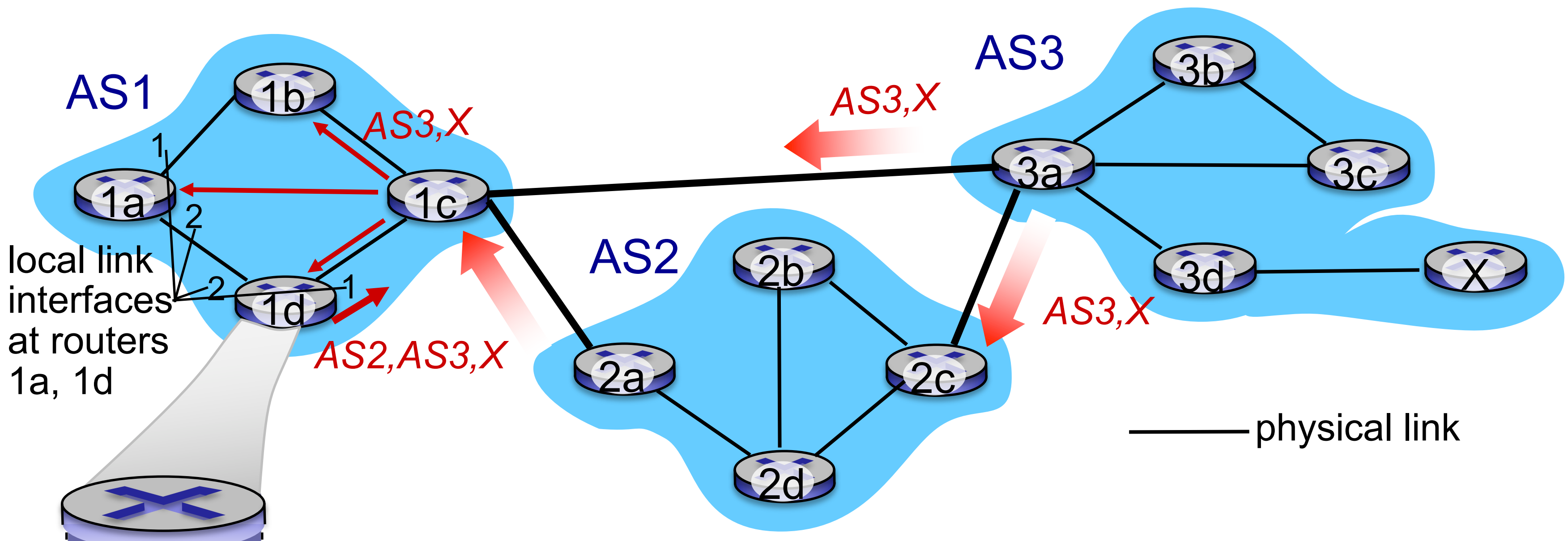


gateway router may learn about multiple paths to destination:

- AS1 gateway router 1c learns path *AS2,AS3,X* from 2a
- AS1 gateway router 1c learns path *AS3,X* from 3a
- Based on policy, AS1 gateway router 1c chooses path *AS3,X, and advertises path within AS1 via iBGP*

BGP, OSPF, forwarding table entries

Q: how does router set forwarding table entry to distant prefix?

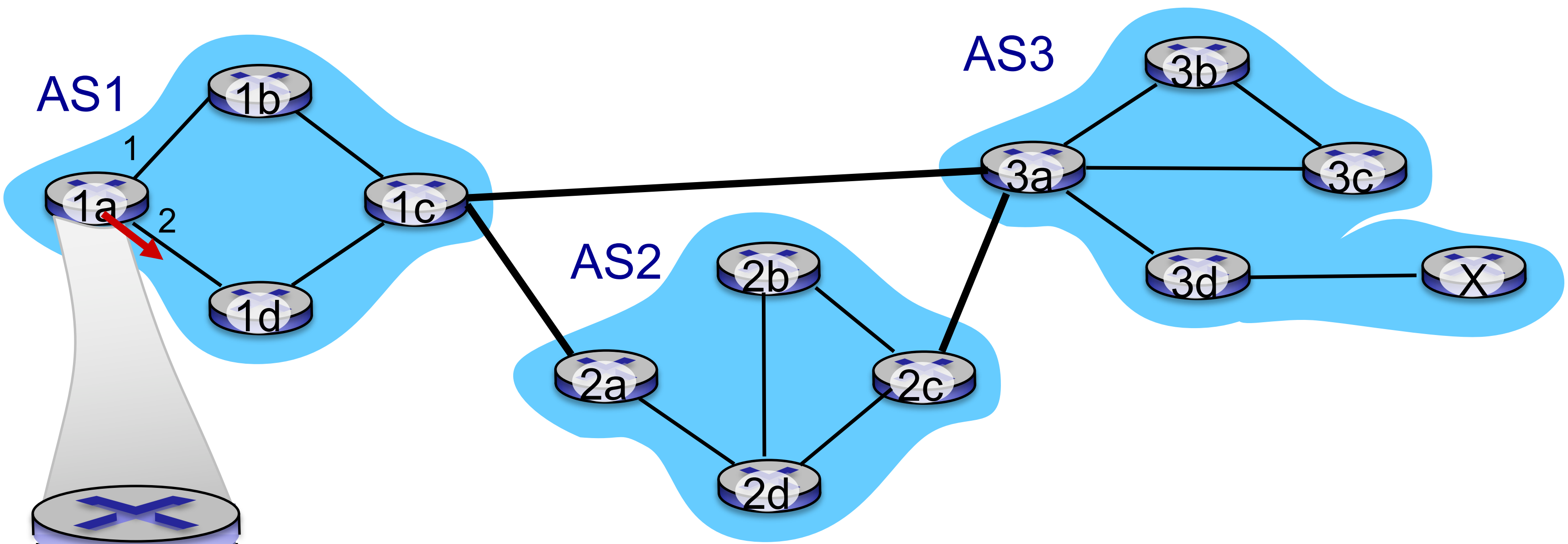


dest	interface
…	…
X	1
…	…

- recall: 1a, 1b, 1d learn about dest X via iBGP from 1c: "path to X goes through 1c"
- **1d**: OSPF intra-domain routing: to get to 1c, forward over outgoing local interface 1

BGP, OSPF, forwarding table entries

Q: how does router set forwarding table entry to distant prefix?

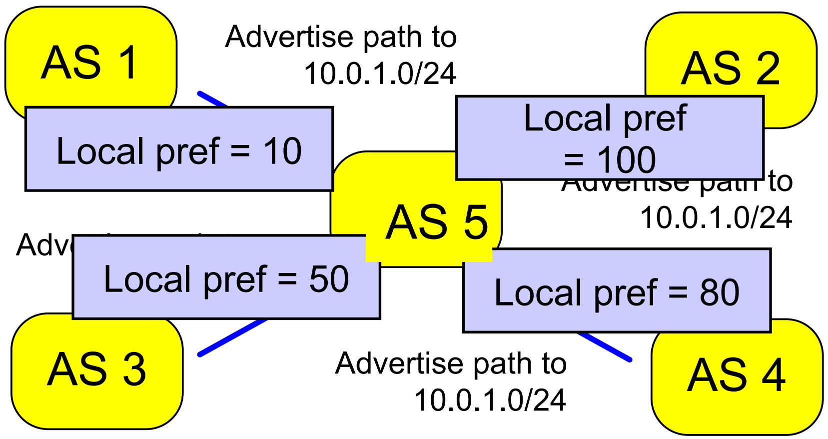


dest	interface
…	…
X	2
…	…

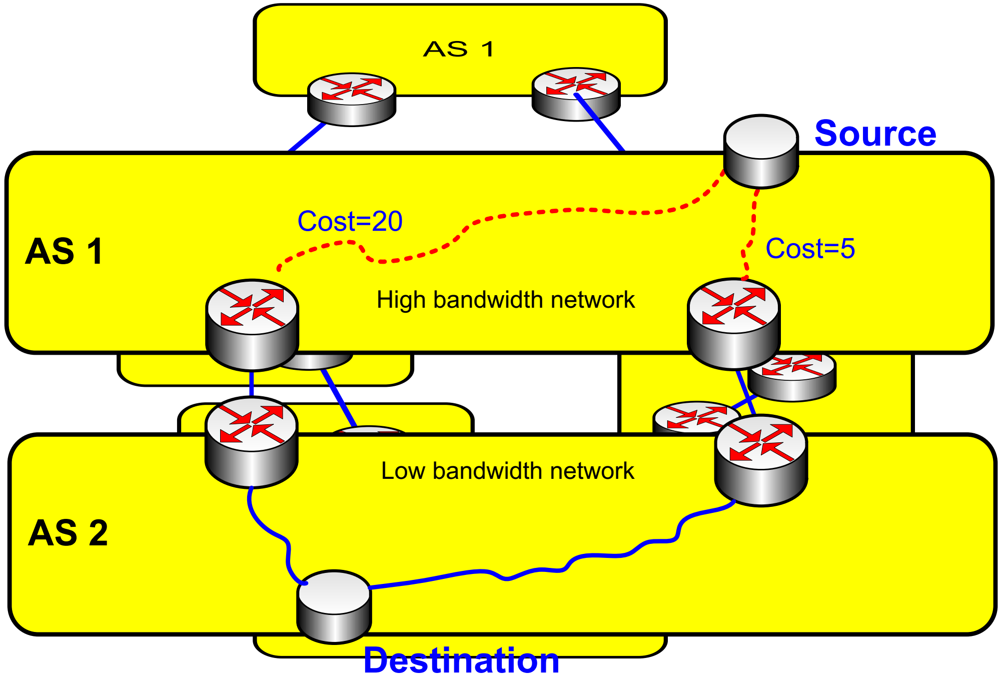
- recall: 1a, 1b, 1d learn about dest X via iBGP from 1c: "path to X goes through 1c"
- 1d: OSPF intra-domain routing: to get to 1c, forward over outgoing local interface 1
- **1a**: OSPF intra-domain routing: to get to 1c, forward over outgoing local interface 2

BGP route selection

- gateway router may learn about more than one route to destination AS, selects route based on:
  1. local preference value attribute: policy decision
  2. shortest AS-PATH
  3. closest NEXT-HOP internal router: hot potato routing
  4. additional criteria

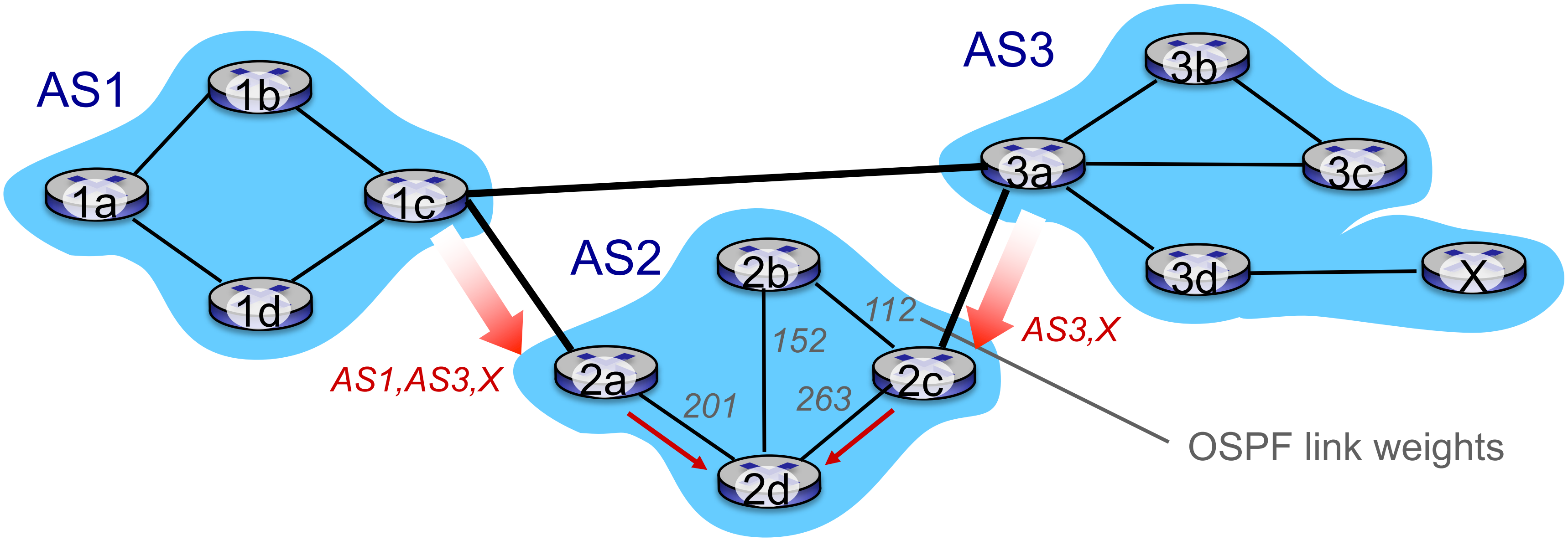


- shortest AS-PATH may not mean shortest router/hop path



- best cost intra path may not mean best cost overall

Hot Potato Routing

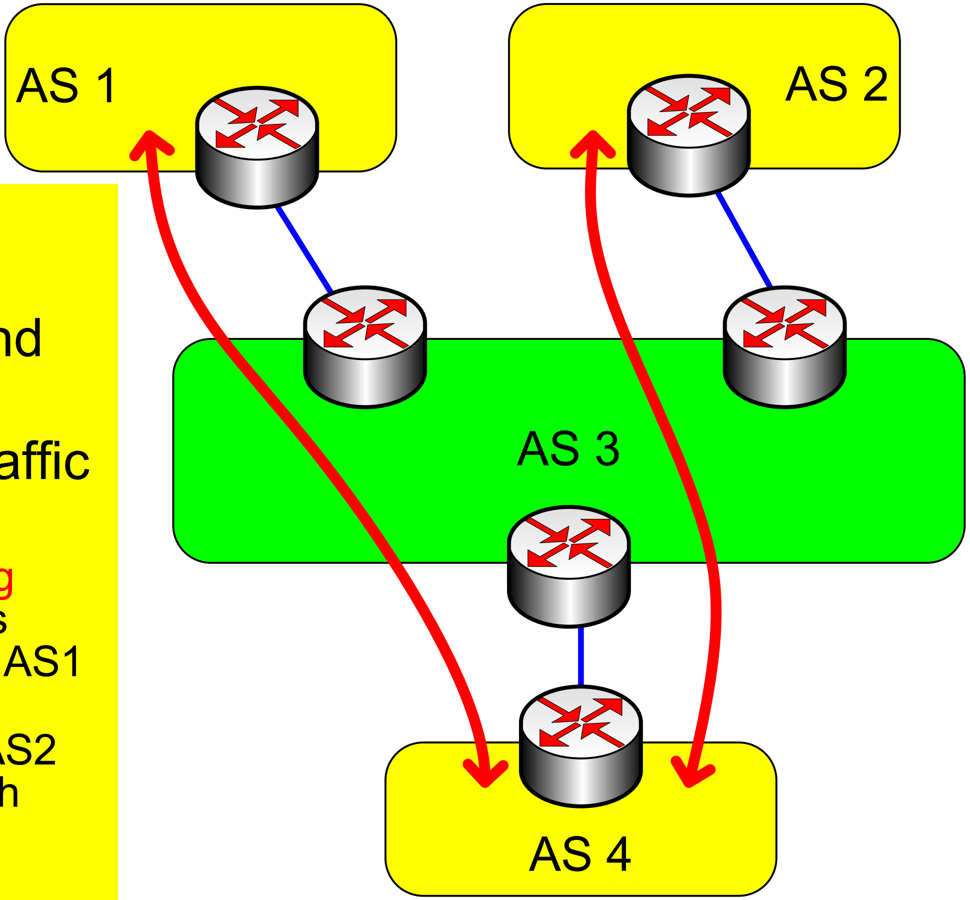


- 2d learns (via iBGP) it can route to X via 2a or 2c
- *hot potato routing:* choose local gateway that has least intra-domain cost (e.g., 2d chooses 2a, even though more AS hops to *X*): don't worry about inter-domain cost!
- Note that BGP will go with shortest AS path first. If 2 routes have same length AS path, use hot potato (shortest path) internally (IGP).

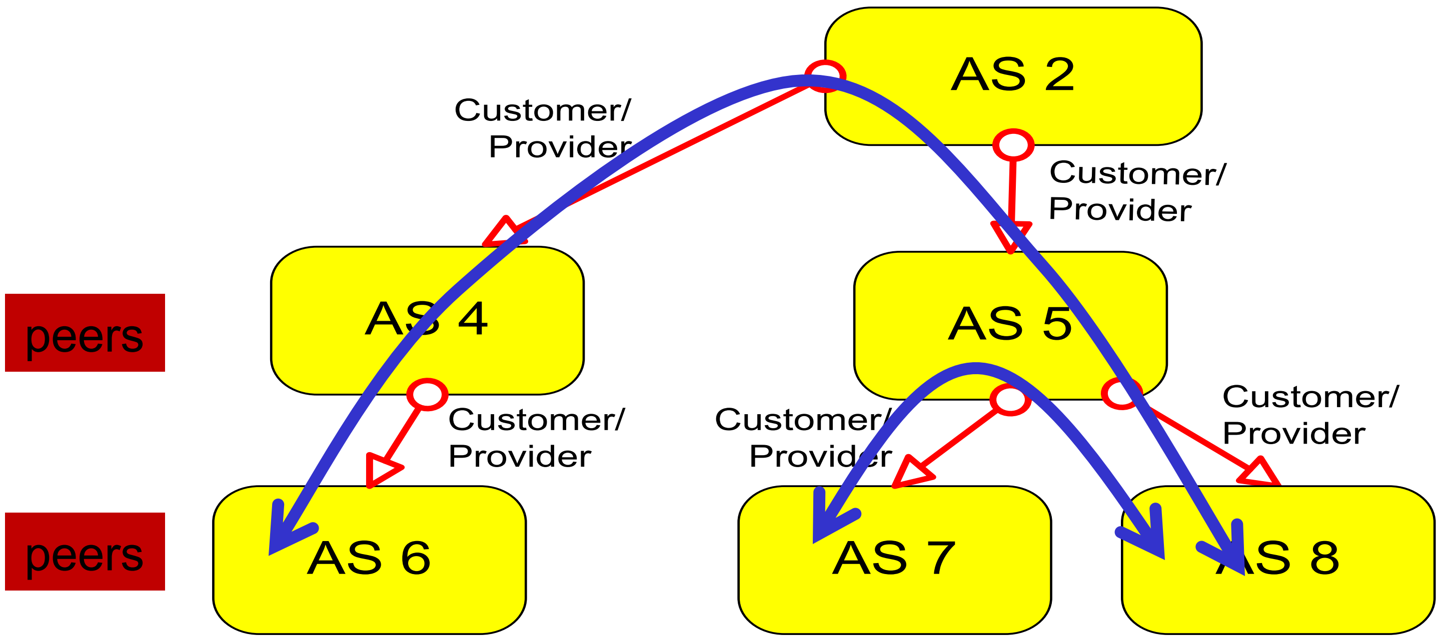
Selective transit

Example:

- AS 3 carries traffic between AS 1 and AS 4 and between AS 2 and AS 4
- But AS 3 does **not** carry traffic between AS 1 and AS 2
    - The example shows a routing policy. In other words, AS3 is perfectly capable of carrying AS1 -> AS2 traffic, but a policy decision prevents AS1 and AS2 from using AS3 to reach each other.



Customer/Provider and Peers



- a stub network typically obtains access to the Internet through a transit network. E.g., AS7 –> <u>AS5</u> –> AS 8
- a transit network that is a provider may be a customer of another network (provider) – AS4 is a customer of AS2 as is AS5.
- customer pays provider for service